

Population, Sample & Sampling Techniques

Tutorial 2

1. What is Population and Sample?

Hello future Data Scientists! Today, we are learning a critical basic concept that everyone analyzing data must know: the difference between a Population and a Sample.

Population: The entire group that we are interested in studying or analyzing.

Sample: A smaller subset (part) taken from the Population.

If you want to extract information from a massive group, asking every single person is usually impossible. It takes too much time and money! Instead, we pick a smaller group (Sample), ask them, and use their answers to guess what the huge group (Population) thinks.

Real-Life Example:

Exit Polls during Elections: Imagine an election is happening in a huge State. The media wants to predict *"Which party will win?"*

- **Population:** Every person in the state who is ≥ 18 years old and has voted. It is impossible to visit every house to ask who they voted for!
- **Sample:** Media reporters visit a few different regions and ask a small number of people who they voted for. Based on this small group's maximum percentage of votes, they predict the winner for the whole state!

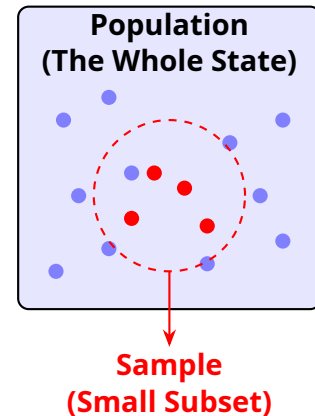
Important Note & Extra Info:

Did you know? Exit polls are based on a sample, which means they are **not always 100% true!** They are just very good mathematical guesses. Also, we can use the average of our sample (like average height) to guess the average of the whole population.

Practice Question:

Q: A chef makes a giant pot of soup. He tastes one spoon to see if it needs more salt. What is the Population and what is the Sample?

A: The whole pot of soup is the **Population**. The one spoon he tastes is the **Sample**!



Topic Summary: Population vs. Sample

The **Population** is the huge overall group we want to study. A **Sample** is a tiny piece of that group we actually collect data from.

2. The Goal of Sampling

Now that we know we need to take a Sample, the big question is: **How do we pick them?** What criteria do we use?

The ultimate goal of sampling is to create a sample that is **representative of the entire population**. This means our small group must accurately reflect the variety present in the huge group. To achieve this, we use different **Sampling Techniques**.

Goal: Make the Sample look just like a mini version of the Population!

3. Types of Sampling Techniques

Type 1: Simple Random Sampling

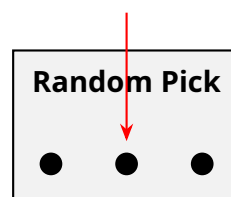
This is the most basic technique! Here, you pick your sample completely at random, without any special conditions.

Important Note & Extra Info:

Golden Rule: When performing Simple Random Sampling, **every member of the population has an equal chance of being selected.**

Real-Life Example:

Putting the names of all 50 students in a class into a hat, mixing them up, and pulling out 5 names blindfolded. Everyone had an equal chance!

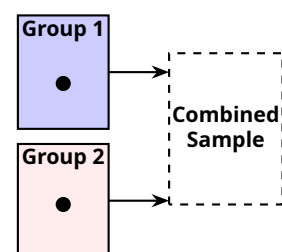


Type 2: Stratified Sampling

If you are a biology student, you might know the word "Strata," which means **layers**. In this technique, we first divide our population into completely separate, **non-overlapping groups** (layers). Then, we pick a random sample from *each* group.

Real-Life Example:

Imagine you want to study household expenses. You divide the population into two non-overlapping groups: **Males** and **Females**. Then, you select 300 Females and 300 Males to ask about expenses (like vegetables, milk, bikes, etc.). By making groups first, you ensure you get answers from everyone!

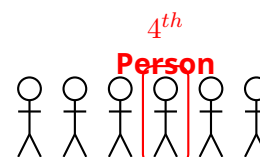


Type 3: Systematic Sampling

Here, we use a system or a math rule to pick people. Instead of picking randomly, we choose every N^{th} **person** from a list or line.

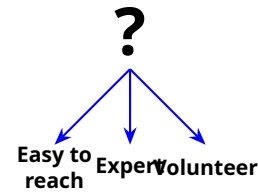
Real-Life Example:

Imagine you are standing outside a shopping mall doing a survey. You decide a system: *"I will ask every 4th person who walks out of the mall to fill out my form!"* You count 1, 2, 3, and ask the 4th person. Then count 1, 2, 3, and ask the 8th person. This is Systematic Sampling!



Type 4: Convenience / Voluntary Response Sampling

Sometimes, we only ask people who are easy to reach or people who are experts in a specific topic. We call this Convenience Sampling or Voluntary Response Sampling.



Real-Life Example:

Expert Survey: If you want to do a survey about "Data Science", you won't ask a doctor, right? You will ask someone who has knowledge of Data Science. You ask experts to *volunteer* to fill the form.

Medical Survey: If you need to know about a specific disease, you will ask Doctors to fill your survey, because it is convenient and they have the exact knowledge.

Practice Question:

Q: You want to know if the school library needs more books. You just ask the 10 students sitting closest to you in class. What type of sampling is this?

A: Convenience Sampling! You picked them simply because they were easy to reach.

Overall Revision Summary

- **Population:** The entire big group of people/items we want to study (e.g., All voters).
- **Sample:** A small portion taken from the population (e.g., A few voters asked in an exit poll).
- **Goal of Sampling:** To create a small group that perfectly represents the big group.
- **Simple Random Sampling:** Everyone has an equal chance of being picked.
- **Stratified Sampling:** Divide into non-overlapping groups (layers), then pick randomly from each group.
- **Systematic Sampling:** Use a rule like "Pick every N^{th} person" (e.g., every 4th person outside a mall).
- **Convenience/Voluntary Sampling:** Pick people who are easy to reach or are experts on the specific topic.

Remember: The domain (topic) decides which sampling technique is best to use! Keep practicing!